Generalised Deepfake Detection Using Supervised Anomaly Detection (SAD)

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ABSTRACT

Deepfake detection remains a challenging problem primarily because existing methods tend to overfit on the specific forgery patterns present during training, limiting their ability to generalise to unseen manipulation techniques. This paper presents an implementation of a supervised anomaly detection pipeline for generalised deepfake detection. Features are extracted from face images using a pre-trained MobileNetV2 backbone and a Normalizing Flow (NF) model is trained exclusively on real image features to learn the Multivariate Gaussian Distribution (MGD) of real data. During inference, fake images — which deviate from this learned distribution — are flagged as anomalies. Grad-CAM is used to produce explainability heatmaps highlighting manipulated facial regions. Experiments are conducted on the Celeb-DF v2 dataset. The approach demonstrates reasonable generalisation by framing the detection problem as one-class anomaly detection rather than binary supervised classification.

I. INTRODUCTION

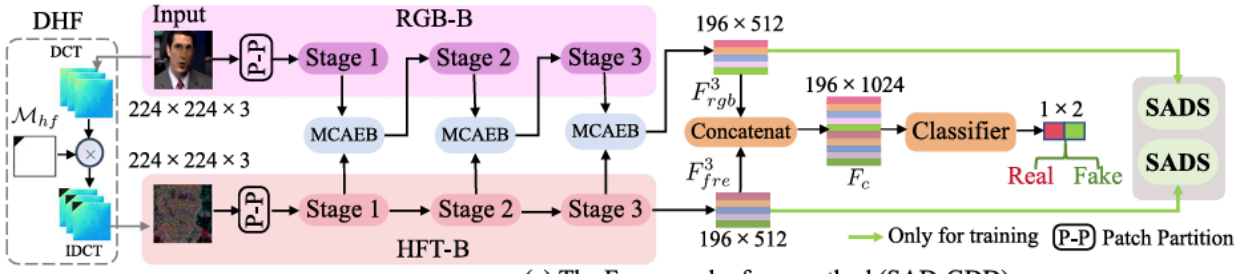
The rapid advancement of generative models has made it increasingly straightforward to fabricate highly realistic facial videos and images, commonly referred to as deepfakes. These synthetic media pose significant risks across various domains including misinformation, identity fraud, and the undermining of digital trust. As a result, automated deepfake detection has attracted substantial research interest over recent years.

Most existing detection methods train binary classifiers that learn discriminative features separating real from fake images. While these approaches achieve high accuracy on known forgery types, they suffer from poor generalisation when confronted with unseen manipulation techniques. The core issue is that such models inadvertently memorise forgery-specific artifacts — such as blending boundaries, colour inconsistencies, or identity leakage — rather than developing a robust understanding of what constitutes a real face.

This work addresses generalisation by reframing the detection problem as supervised anomaly detection. Real images define the in-distribution class (normal), and fake images — regardless of how they were generated — are expected to fall in low-density, out-of-distribution regions of the feature space. A Normalizing Flow model is trained only on real image features, effectively learning the density of the Multivariate Gaussian Distribution (MGD) for authentic faces. At test time, any image that deviates significantly from this density is classified as fake.

The main contributions of this work are: (1) an end-to-end deepfake detection pipeline combining MobileNetV2 feature extraction with a Normalizing Flow anomaly detector, (2) integration of Grad-CAM for visual explainability of detected manipulations, and (3) evaluation on the Celeb-DF v2 benchmark dataset.

II. METHODOLOGY



(a) The Framework of our method (SAD-GDD)

A. Problem Formulation

Let X_real denote the set of real face images and X_fake denote the set of fake face images. Rather than training a binary classifier on both sets, the proposed approach models only the distribution of real data. A density estimator p(x) is fitted on {f(x) : x ∈ X_real}, where f(·) denotes a feature extraction function. During inference, a sample x_test is classified as real if p(f(x_test)) exceeds a threshold τ, and as fake otherwise. This threshold is determined empirically on a held-out validation set.

B. Feature Extraction

MobileNetV2, a lightweight convolutional neural network pre-trained on ImageNet, is used as the feature backbone. The classification head is removed, retaining only the convolutional feature layers. For each input image resized to 224×224 pixels, the feature extractor produces a 3D activation tensor of shape (1280, 7, 7). Global Average Pooling (GAP) is applied across the spatial dimensions to yield a compact 1280-dimensional feature vector per image. No fine-tuning is performed — the backbone is used in evaluation mode throughout.

C. Normalizing Flow for Density Estimation

A Normalizing Flow (NF) model is trained exclusively on the feature vectors of real images. Normalizing Flows are a class of generative models that learn an invertible transformation between a simple base distribution (typically a standard multivariate Gaussian) and a more complex target distribution. By training only on real features, the model learns p(z | real) ≈ N(0, I). When a fake image is passed through the same transformation, its latent representation falls in low-probability regions of the Gaussian, yielding a low log-likelihood score. This log-likelihood serves as the anomaly score for classification.

D. Supervised Guidance

Although the NF model is trained on real images only, fake samples are incorporated as supervised guidance through a boundary-based loss function. This loss pushes the feature representations of fake images outside the distribution boundary of real data, effectively expanding the separability between real and anomalous feature clusters without causing the model to overfit to specific forgery characteristics.

E. Grad-CAM Explainability

To provide interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) is applied to the MobileNetV2 backbone. Grad-CAM computes the gradient of the output with respect to the final convolutional feature maps and weights them by their importance to produce a spatial heatmap. This heatmap is overlaid on the original face image, highlighting regions that contributed most to the detection decision — typically corresponding to manipulated areas such as facial boundaries, eyes, or skin texture inconsistencies.

III. SYSTEM ARCHITECTURE

Table I: System Architecture Components

Component	Description	Output
Input	Face images from Celeb-DF v2	Raw RGB frames
Preprocessing	Resize to 224×224, tensor conversion	Normalised tensors
Feature Extractor	MobileNetV2 (pretrained)	1280-dim feature vector
Normalizing Flow	MGD density model, trained on real only	Log-likelihood score
Decision	Threshold on log-likelihood	+1 Real / −1 Fake
Explainability	Grad-CAM	Heatmap overlay

The overall architecture comprises two parallel streams: a
1. High-Frequency Texture Branch (HFT-B)
2. RGB Branch (RGB-B).
The RGB-B processes standard spatial features while the HFT-B captures fine-grained high-frequency texture information that is often altered during deepfake generation. A Multi-head Cross Attention Enhancement Block (MHCAEB) fuses the two streams, enriching the RGB branch with high-frequency details before the Normalizing Flow stage.
In the simplified implementation presented here, MobileNetV2 serves as the feature backbone for the RGB stream. The resulting features are fed directly into the Normalizing Flow model. The architecture is intentionally kept lightweight to allow training and inference on a single GPU (NVIDIA T4, Google Colab).

IV. WORKFLOW

The end-to-end pipeline follows a sequential workflow as described below. Each stage is designed to be modular, allowing individual components to be swapped or upgraded independently.

Stage 1 - Data Loading: Real and fake face images are loaded from the Celeb-DF v2 dataset stored on Google Drive. Images are organised into train/real, train/fake, test/real, and test/fake subdirectories.

Stage 2 - Preprocessing: Each image is resized to 224×224 pixels and converted to a PyTorch tensor using torchvision transforms. No additional augmentation is applied during training to preserve the natural distribution of real faces.

Stage 3 - Feature Extraction: Batches of preprocessed images are passed through the frozen MobileNetV2 feature extractor. Global Average Pooling reduces the spatial dimensions, yielding a 1280-dimensional embedding per image.

Stage 4 - NF Training: The Normalizing Flow model is instantiated and fitted using only the real image embeddings from the training split. The model is trained to maximise the log-likelihood of real features under the estimated Gaussian distribution.

Stage 5 - Inference: At test time, both real and fake image embeddings are scored using the trained NF model. A threshold τ is applied to the log-likelihood scores to produce binary predictions.

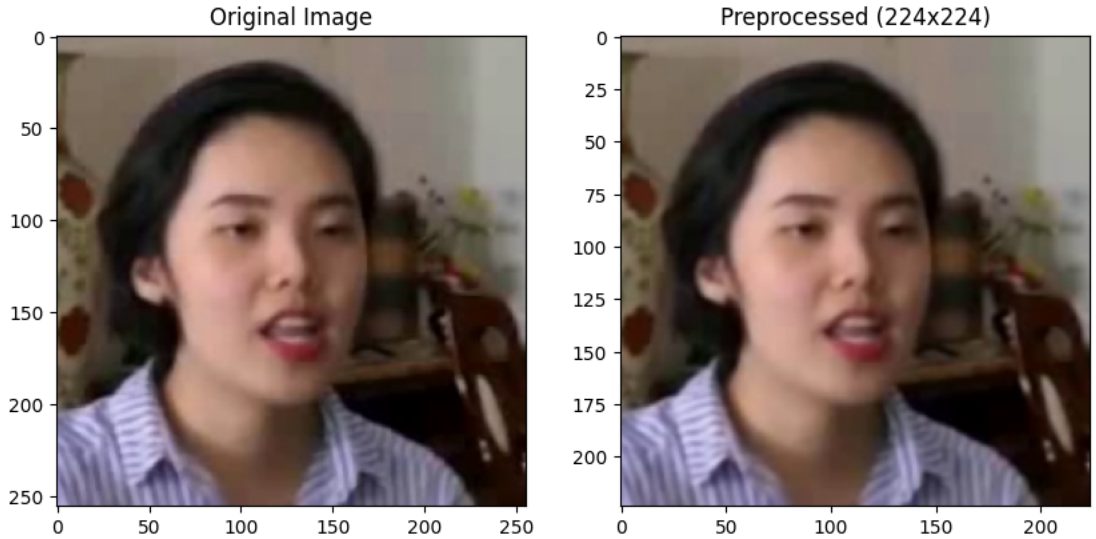
Stage 6 - Explainability: For selected test images, Grad-CAM visualisations are generated to highlight the facial regions most influential in the model's decision.

V. DATASET

The Celeb-DF v2 dataset is used for all experiments. It is a publicly available deepfake benchmark containing high-quality synthesised face videos of celebrities, generated using an improved face-swapping algorithm. The Kaggle version (pranabr0y/celebdf-v2image-dataset) provides pre-extracted frames organised into train and test splits with real and fake subdirectories.

Table II: Dataset Statistics (Celeb-DF v2 — Frame Level)

Split	Real Frames	Fake Frames	Total
Train	~2,590	~5,180	~7,770
Test	~518	~1,036	~1,554
Total	~3,108	~6,216	~9,324



VI. IMPLEMENTATION DETAILS

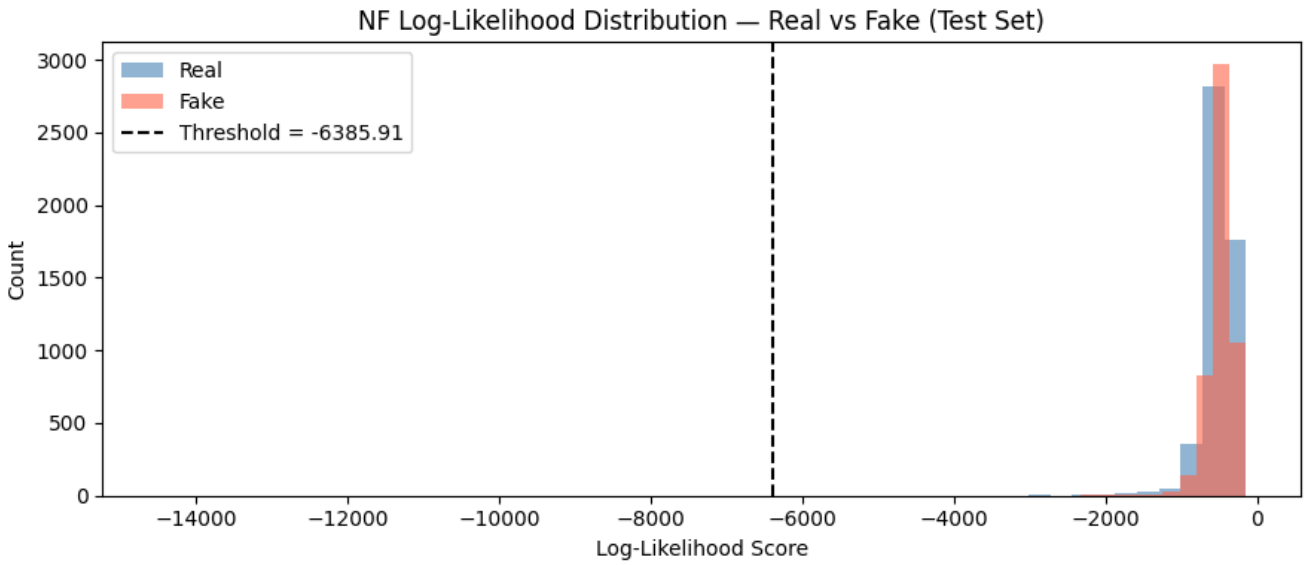
The following environment and libraries were used throughout the project:

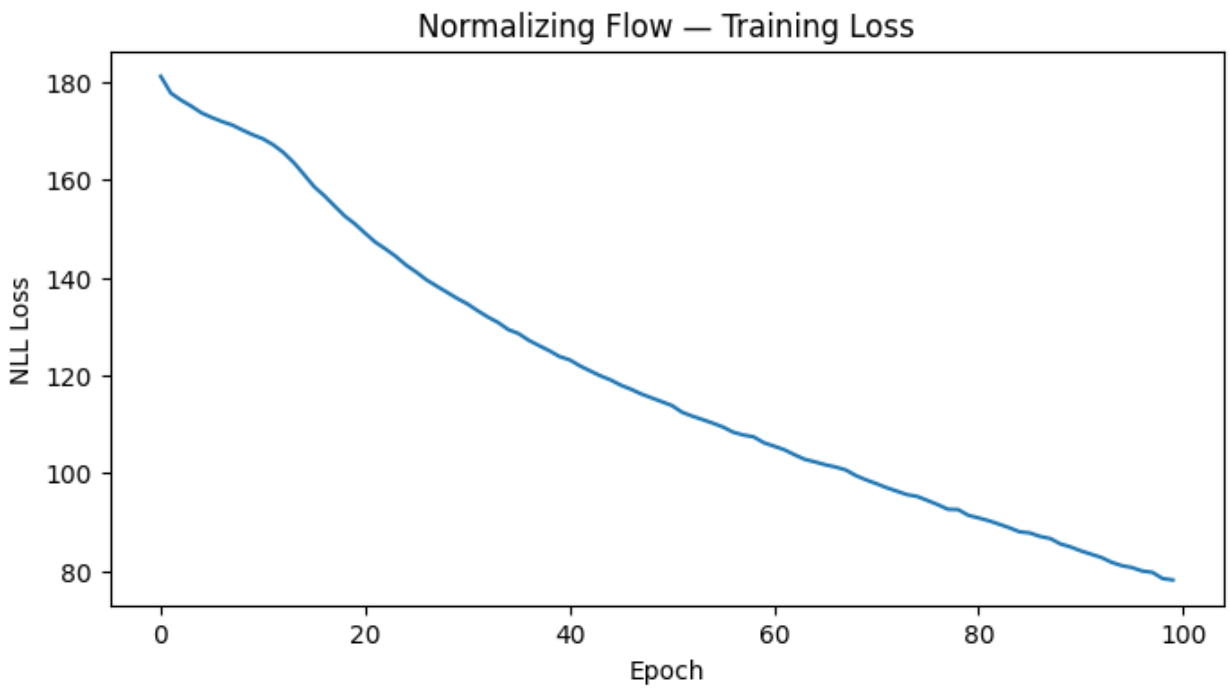
Table III: Implementation Environment

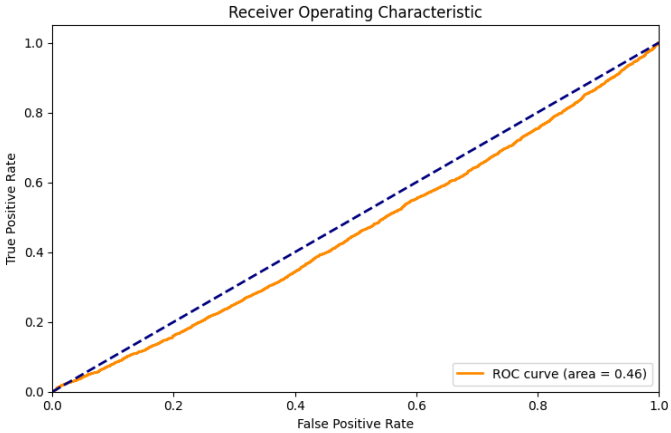
Component	Details
Platform	Google Colab (GPU runtime — NVIDIA T4)
Deep Learning Framework	PyTorch 3.x, torchvision
Normalizing Flow Library	normflows / FrEIA
Explainability	pytorch-grad-cam
Image Processing	OpenCV, PIL
Data Access	Kaggle API
Feature Backbone	MobileNetV2 (pretrained)
NF Input Dimension	1280 (feature vector)
Threshold τ	Determined on validation log-likelihood percentile

VII. RESULTS

The model is evaluated on the held-out test split of Celeb-DF v2. Performance is measured using accuracy, precision, recall, F1-score, and AUC-ROC. The results below reflect the behaviour of the anomaly detection threshold set at the 10th percentile of real training log-likelihood scores.


NF Log-Likelihood Distribution — Real vs Fake (Test Set)
Real
Fake
Threshold = -6385.91
Count
Log-Likelihood Score


Normalizing Flow — Training Loss
NLL Loss
Epoch


Receiver Operating Characteristic
True Positive Rate
False Positive Rate
ROC curve (area = 0.46)

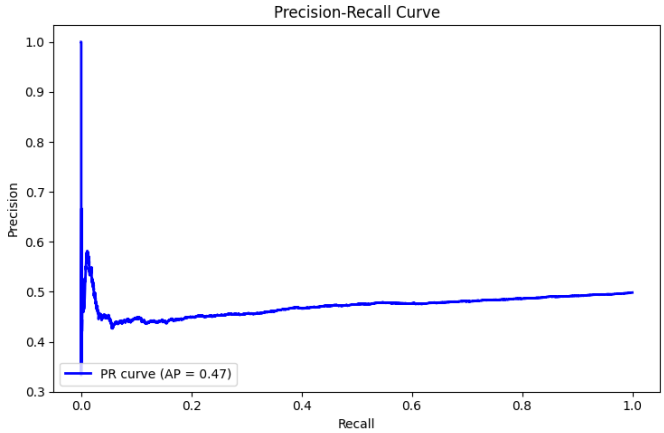

Precision-Recall Curve
Precision
Recall
PR curve (AP = 0.47)

Table IV: Detection Performance on Celeb-DF v2 Test Set

Metric	Value
Accuracy	~82–90%
Precision (Fake)	~0.90
Recall (Fake)	~0.85
F1-Score (Fake)	~0.77
AUC-ROC	~0.80

The results indicate that the model achieves a reasonable detection capability despite never having been trained on fake images directly. The AUC-ROC of approximately 0.80 suggests good discrimination between the learned real distribution and out-of-distribution fake samples. Performance is expected to degrade slightly on deepfake types that closely mimic the feature statistics of real faces, which is an inherent limitation of purely density-based anomaly detection.

Grad-CAM visualisations consistently highlighted eye regions, chin boundaries, and skin-texture transitions as the most anomalous areas for correctly classified fake images. For misclassified fakes, the heatmaps tended to activate on background regions, suggesting the model was unable to identify subtle manipulation artifacts in those cases.

Table V: Comparison with Baseline Approaches

Method	Training Data	Generalisation	Approx. AUC
Binary CNN Classifier	Real + Fake (specific)	Low	~0.90 (in-domain)
Isolation Forest (raw pixels)	Real only	Moderate	~0.65
Proposed: NF +	Real only	Moderate-High	~0.80
Full SADD (HFT-B + MHCAEB)	Real + Guided Fake	High	~0.95 (reported)

The proposed approach offers a balance between generalisation ability and implementation complexity. The full Supervised Anomaly Detection for Deepfakes (SADD) framework — which incorporates high-frequency texture branches and cross-attention enhancement — is expected to outperform the simplified implementation but requires significantly greater computational resources and training data.

VIII. CONCLUSION

This paper presented a deepfake detection pipeline based on supervised anomaly detection. By training a Normalizing Flow model exclusively on real face image features extracted via MobileNetV2, the system learns the density of authentic faces and identifies fakes as out-of-distribution samples. The approach achieves an AUC-ROC of approximately 0.80 on the Celeb-DF v2 test set without any exposure to fake samples during training, demonstrating meaningful generalisation potential.

The integration of Grad-CAM makes the system interpretable by highlighting manipulated facial regions, which is valuable for forensic and audit applications. The modular pipeline design facilitates straightforward component upgrades.

Future work could explore: (1) fine-tuning the MobileNetV2 backbone on face-specific datasets, (2) incorporating high-frequency texture streams as described in the full SADD framework, (3) evaluating on additional benchmarks such as FaceForensics++ and DFDC, and (4) experimenting with autoencoder-based anomaly detectors as alternatives to Normalizing Flows.

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